

A Run Expectancy Approach to Lead Distance Optimization in Major League Baseball

Zach Sissman¹, Lila Dodson², Jack Whitney-Epstein³, and Jonathan Pipping-Gamón⁴

¹Community School of Naples, Naples, FL

²San Francisco University High School, San Francisco, CA

³The Brunswick School, Greenwich, CT

⁴Department of Statistics & Data Science, University of Pennsylvania

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Abstract

In baseball, a runner’s primary lead off first base creates a trade-off: a larger lead improves the runner’s chance of stealing second but also increases exposure to pickoff attempts. Using 2024 MLB pitch-level data and Baseball Savant tracking measures, we model this trade-off as a sequential process with four stages: pickoff attempt, pickoff success conditional on an attempt, steal attempt conditional on no pickoff, and steal success conditional on an attempt. Each stage is estimated with logistic or mixed-effects logistic regression using lead distance, pitcher hold quality, runner sprint speed, catcher pop time, disengagement state, and outs, with runner and catcher random effects where appropriate. We then map the fitted stage probabilities to outs-specific RE24 run values and choose the lead distance that maximizes expected runs for each runner–pitcher–catcher and game context. Observed leads are slightly larger than the model-implied optimum on average: +0.13 ft across the analyzed runner-on-first opportunities and +0.10 ft on steal attempts. Translating these deviations to a common run-expectancy scale, we estimate a model-implied total of 38.8 expected runs lost across the analyzed 2024 sample and find that aggressive deviations carry roughly 3.3 times the per-opportunity run cost of conservative deviations. The framework converts the steal–pickoff trade-off into a common run-expectancy scale and provides context-specific lead estimates.

1 Introduction

1.1 History of Base Stealing

In baseball, scouts traditionally evaluate five tools: hitting for average, hitting for power, fielding, throwing, and speed. Among these, speed is often undervalued, despite playing a

central role in a player’s effectiveness on the basepaths. Stolen bases are the most visible expression of that impact.

Historically, steals have waxed and waned with MLB’s run-scoring environment. During the Deadball era (c. 1900–1920), home run rates were low and teams relied more on advancing runners through stolen bases. Many seasons saw clubs exceed 100 steals and leagues tally well over 1,000 steals combined (McMurray, 2015). As home runs rose from the 1930s through the 1950s, average team steals fell to roughly 39 per season (Baseball-Reference, nd). From the 1960s to the 1980s, players such as Lou Brock and Rickey Henderson brought base-stealing back into prominence. Maury Wills’s 1962 season, which included 104 stolen bases, was a turning point in baserunning and influenced how opponents pitched and held runners on (Vazzana, 2016).

1.2 Sabermetrics and Expected Runs

By the 1990s and 2000s, sabermetrics reframed the value of the steal. Bill James argued that a steal must succeed about two-thirds of the time to break even, a rule of thumb that recognizes outs as scarce resources and emphasizes efficiency over raw totals (James, 2023). Building on this logic, linear weights assign average run values to outcomes independent of specific context. On Baseball Savant, a successful steal is valued at +0.20 expected runs, while being caught stealing or picked off is valued at -0.45 expected runs (Baseball Savant, nd). These values provide a useful baseline for evaluation.

However, context still matters when deciding whether to steal. The run value of a steal depends on the base-out state, defined by which bases are occupied and the number of outs. There are $2^3 \times 3 = 24$ possible base-out states. Sabermetrics defines the value of a play as the change in run expectancy between states, and run expectancy tables such as Table 1 are commonly used to estimate this value. According to Table 1, stealing second base with 0 outs and no other runners on base is worth $1.068 - 0.831 = 0.237$ runs, while being caught stealing in the same situation is worth $0.243 - 0.831 = -0.588$ runs (FanGraphs, nd).

Table 1: Example run expectancy table

Runners	0 Outs	1 Out	2 Outs
Empty	0.461	0.243	0.095
1 _ _	0.831	0.489	0.214
_ 2 _	1.068	0.644	0.305
1 2 _	1.373	0.908	0.343
_ _ 3	1.426	0.865	0.413
1 _ 3	1.798	1.140	0.471
_ 2 3	1.920	1.352	0.570
1 2 3	2.282	1.520	0.736

Example run expectancy table (FanGraphs, nd).

1.3 The Lead’s Impact on Base Stealing

While speed is a key ingredient in stealing bases, the lead a runner takes off first base also shapes both steal success and pickoff risk. A larger lead shortens the distance to second if the runner goes, but it also increases exposure to pickoff attempts at first.

This strategic dimension of stealing has been studied in game-theoretic terms. [Turocy \(2014\)](#) models the steal decision as a two-player inspection game between the runner and the defense, where the runner chooses whether to go and the defense allocates attention between the runner and the batter. The mixed-strategy equilibrium implies a near-linear relationship between attempt frequency and success rate, with validation over MLB play-by-play data from 1974–2011. However, the framework does not prescribe an optimal primary lead or incorporate pitcher- and catcher-specific skills at the pitch level.

A complementary physics approach examines the kinematics of a steal attempt. Using position–time and velocity–time profiles for a case study, [Kagan \(2013\)](#) parametrizes acceleration from first, slide deceleration, top speed, speed upon arrival, and the runner’s lead distance. He finds that sprint speed and initial acceleration most strongly affect success, with comparatively modest effects attributed to the primary lead. This analysis is mechanically useful, but it only implicitly considers pickoff risk and is limited in scope.

Data-driven work using tracking-era measurements shows how player behavior and pitcher tendencies shape leads. [Lindbergh \(2015\)](#) uses MLB data on primary and secondary leads and sprint speed, highlighting outliers such as Jon Lester’s reluctance to throw over and Ichiro Suzuki’s larger-than-expected leads relative to speed. That study also notes a positive association between sprint speed and lead size. It does not, however, jointly model runner, pitcher, and catcher characteristics, nor does it connect the full decision sequence from pickoff attempt to steal outcome to expected runs.

More recently, [Powers \(2026\)](#) models lead distance under MLB’s 2023 disengagement-limit rules as a sequential zero-sum game between the runner and pitcher. Using generalized linear mixed-effects models and a Markov decision process, Powers estimates that optimal lead-distance policies vary by count, outs, and prior pickoff attempts. Powers also finds that runners should substantially extend their leads after disengagements, while pitchers attempt fewer pickoffs than the game-theoretic optimum. However, that framework focuses primarily on equilibrium strategies under league-average conditions rather than estimating a context-specific optimal lead for individual runner–pitcher–catcher matchups on a unified expected-runs scale.

This study extends the existing literature in four ways. First, we model the baserunning sequence with a nested set of logistic regressions. Second, we incorporate runner sprint speed, pitcher hold ability (*Threat*), catcher pop time, disengagement state, and outs in the relevant stages. Third, we use random effects to account for unobserved runner and catcher heterogeneity. Fourth, we optimize primary lead distance to maximize expected runs for the base state with a runner on first, with second and third base empty. In doing so, we quantify the reward of a larger lead and the cost of greater pickoff risk on a common run-expectancy scale.

2 Methodology

2.1 Data Overview

Our primary dataset, provided by MLB, consists of 2024 pitch-level observations. It includes all pickoff attempts, a random sample of called pitches, and every stolen-base attempt on takes, defined as pitches where the batter did not swing. The unit of observation is a single pitch. Each observation includes pitch context (inning, date, home/away, count, and outs), participant IDs (runner, pitcher, and catcher), and the runner’s primary and secondary lead distances in feet. A primary lead is defined as the distance from the runner’s center of mass to the respective base at the time of the pitcher’s first movement. We denote the primary lead by L_i . To avoid base-state confounding, we restrict the analysis to the state with a runner on first, with second and third base empty. Under this restriction, the only base-out component that varies in the run-expectancy calculation is the number of outs.

We merge the pitch-level data with the following covariates from Baseball Savant:

- Runner sprint speed, v_i (feet per second).
- Catcher pop time, p_i (seconds), defined as the average time it takes a catcher to throw the ball to second base after receiving a pitch.
- Pitcher *Threat*, θ_i (per 100 IP): Baseball Savant’s Net Bases Prevented ([Baseball Savant](#), [nd](#)) scaled to 100 innings pitched, $\theta_i = 100 \text{ NBP/IP}$. Higher values indicate stronger control of the running game.
- Disengagement state, $r_i \in \{0, 1, 2\}$, an indicator capturing the 2023 MLB rule limiting pickoff attempts. Under this rule, a pitcher is allowed two pickoffs or step-offs per plate appearance; a third disengagement results in a balk unless the runner is successfully picked off. Since we do not observe step-offs, we construct this variable using only observed pickoff attempts: $r_i = 0$ indicates no prior pickoffs, $r_i = 1$ indicates one prior pickoff, and $r_i = 2$ indicates two or more prior pickoff attempts.
- Outs, $o_i \in \{0, 1, 2\}$.

All continuous predictors (lead distance, pitcher Threat, catcher pop time, and runner sprint speed) are standardized by subtracting their sample mean and dividing by their sample standard deviation. This places predictors on comparable scales. In the logistic models below, each continuous coefficient represents the change in log-odds associated with a one-standard-deviation increase in the predictor. In the optimization step, candidate lead distances are evaluated in feet but standardized using the same lead-distance mean and standard deviation used to fit the models; the resulting optimum is then transformed back to feet for reporting.

To evaluate the expected value of base-stealing outcomes, we use situational weights derived from the RE24 run expectancy table (Table 1):

$$\text{xRuns}_i = w_{\text{SB}}(o_i) \cdot \mathbf{1}\{\text{SB}_i\} + w_{\text{CS}}(o_i) \cdot \mathbf{1}\{\text{CS}_i\} + w_{\text{PK}}(o_i) \cdot \mathbf{1}\{\text{PK}_i\}. \quad (1)$$

Here $w_{\text{SB}}(o_i)$, $w_{\text{CS}}(o_i)$, and $w_{\text{PK}}(o_i)$ are the changes in run expectancy associated with a stolen base, caught stealing, and pickoff, respectively, given the current number of outs.

Because the analysis is restricted to a runner on first, with second and third base empty, these weights vary only by outs.

2.2 Modeling Framework

We represent the runner-on-first sequence as a set of conditional events. First, the pitcher may attempt a pickoff. Conditional on a pickoff attempt, the runner may be picked off. If no pickoff is attempted, the runner may attempt to steal. Conditional on a steal attempt, the steal may succeed. Pitches on which none of these events occur are classified as no-event pitches and receive zero incremental run value in Equation 1.

We model the pickoff-attempt stage with a fixed-effects logistic regression and the pickoff-success, steal-attempt, and steal-success stages with mixed-effects logistic regressions. Random intercepts are included for runners in the pickoff-success stage and for both runners and catchers in the steal-attempt and steal-success stages. Outs enters the pickoff-attempt, steal-attempt, and steal-success stages directly as a predictor; it also enters the expected-runs objective through the outs-specific RE24 weights. Formally,

$$\mathbb{P}(\text{PO}_i \mid L_i, \theta_i, r_i, o_i) = \text{logit}^{-1}(\alpha_0 + \alpha_1 L_i + \alpha_2 \theta_i + \alpha_3 r_i + \alpha_4 o_i), \quad (2)$$

$$\mathbb{P}(\text{PK}_i \mid \text{PO}_i, L_i, \theta_i, r_i) = \text{logit}^{-1}(\beta_0 + \beta_1 L_i + \beta_2 \theta_i + \beta_3 r_i + u_{i,\text{PK}}), \quad (3)$$

$$\mathbb{P}(\text{ATT}_i \mid \neg \text{PO}_i, \theta_i, p_i, v_i, r_i, o_i) = \text{logit}^{-1}(\gamma_0 + \gamma_1 \theta_i + \gamma_2 p_i + \gamma_3 v_i + \gamma_4 r_i + \gamma_5 o_i + u_{i,\text{ATT}} + z_{i,\text{ATT}}), \quad (4)$$

$$\mathbb{P}(\text{SB}_i \mid \text{ATT}_i, L_i, \theta_i, p_i, v_i, r_i, o_i) = \text{logit}^{-1}(\delta_0 + \delta_1 L_i + \delta_2 \theta_i + \delta_3 p_i + \delta_4 v_i + \delta_5 r_i + \delta_6 o_i + u_{i,\text{SB}} + z_{i,\text{SB}}). \quad (5)$$

Here $u_{i,\cdot}$ and $z_{i,\cdot}$ denote runner- and catcher-specific random intercept terms for the corresponding stages. The notation r_i and o_i in Equations 2–5 is shorthand for categorical fixed effects, with 0 prior disengagements and 0 outs used as the reference levels.

We exclude outs from the pickoff-success-given-attempt stage. Once a throw-over occurs, pickoff success is more directly tied to lead distance, pitcher control of the running game, runner traits, and disengagement context than to the base-out valuation. This restriction also keeps the pickoff-success stage parsimonious, since successful pickoffs are relatively rare.

We also exclude primary lead distance from the steal-attempt model. This treats the attempt model as capturing the runner’s baseline steal-attempt propensity after conditioning on pitcher, catcher, runner, disengagement state, and outs. Including lead distance in this stage would risk using the runner’s realized pre-pitch positioning as a proxy for unobserved steal intent, which would partially absorb the effect of lead distance that the optimization is designed to study.

Let $X_i = (\theta_i, p_i, v_i, r_i, o_i)$ denote the observed covariate context for pitch i . When player-specific random effects are used in prediction, the full prediction context also includes runner

and catcher IDs. The unconditional stage probabilities are then

$$\begin{aligned}\mathbb{P}(\text{PK}_i \mid L_i, X_i) &= \mathbb{P}(\text{PK}_i \mid \text{PO}_i, L_i, \theta_i, r_i) \cdot \mathbb{P}(\text{PO}_i \mid L_i, \theta_i, r_i, o_i), \\ \mathbb{P}(\text{ATT}_i \mid L_i, X_i) &= \mathbb{P}(\text{ATT}_i \mid \neg\text{PO}_i, \theta_i, p_i, v_i, r_i, o_i) \cdot (1 - \mathbb{P}(\text{PO}_i \mid L_i, \theta_i, r_i, o_i)), \\ \mathbb{P}(\text{SB}_i \mid L_i, X_i) &= \mathbb{P}(\text{SB}_i \mid \text{ATT}_i, L_i, X_i) \cdot \mathbb{P}(\text{ATT}_i \mid L_i, X_i), \\ \mathbb{P}(\text{CS}_i \mid L_i, X_i) &= (1 - \mathbb{P}(\text{SB}_i \mid \text{ATT}_i, L_i, X_i)) \cdot \mathbb{P}(\text{ATT}_i \mid L_i, X_i).\end{aligned}$$

Taking conditional expectations of Equation 1 gives

$$\begin{aligned}\mathbb{E}[\text{xRuns}_i \mid L_i, X_i] &= w_{\text{SB}}(o_i) \cdot \mathbb{P}(\text{SB}_i \mid L_i, X_i) + w_{\text{CS}}(o_i) \cdot \mathbb{P}(\text{CS}_i \mid L_i, X_i) \\ &\quad + w_{\text{PK}}(o_i) \cdot \mathbb{P}(\text{PK}_i \mid L_i, X_i).\end{aligned}\tag{6}$$

Thus, the expected run value of a candidate lead is estimable from the four fitted stage models and the outs-specific RE24 weights.

2.3 Objective and Optimization

For each fixed context $X_i = (\theta_i, p_i, v_i, r_i, o_i)$, we evaluate the expected run value of candidate primary leads using Equation 6. The model-implied optimal lead is

$$L_i^* = \arg \max_{L \in \mathcal{L}} \text{xRuns}(L \mid X_i),\tag{7}$$

where $\mathcal{L}$ denotes the candidate lead range. In implementation, $\mathcal{L} = [0, 90]$ is used as a broad numerical bound for optimization, not as a physically meaningful range of feasible primary leads. Substantive interpretations are restricted to optima that fall within the empirically relevant range of observed primary leads.

We solve Equation 7 using R’s `optimize()` function, which implements Brent’s method (Brent, 1973). Because any optimum selected near the boundary of the candidate range would depend heavily on extrapolation, such cases should be treated as diagnostic rather than as actionable recommendations. We compare observed leads to L_i^* to summarize how far runners are from the model-implied optimum.

3 Results

3.1 Model Estimates

Table 2 summarizes the main fitted-model patterns; full coefficient estimates are reported in Appendix Tables 3–6. All interpretations below are associative rather than causal, conditional on the included covariates and model specification. The fitted models recover the expected qualitative trade-off: larger leads are associated with both more pickoff attempts and substantially higher pickoff success conditional on an attempt, while also being associated with higher steal success conditional on a steal attempt.

In the pickoff-attempt model (Table 3), a one-standard-deviation increase in lead distance is associated with roughly 36% higher odds of a throw-over ($\exp(0.309) \approx 1.36$), while a

Table 2: Summary of fitted model patterns

Stage	Main fitted pattern	Random effects	Obs.
PO	Larger leads are associated with higher pickoff-attempt odds; prior disengagements also increase attempt odds.	None	72,645
PK PO	Larger leads and stronger pitcher <i>Threat</i> are associated with higher pickoff-success odds.	Runner	7,439
ATT ¬PO	Higher runner sprint speed and prior disengagements increase steal-attempt odds; stronger pitcher <i>Threat</i> lowers attempt odds.	Runner, catcher	65,206
SB ATT	Larger leads and slower catcher pop times are associated with higher steal-success odds.	Runner, catcher	2,434

one-standard-deviation increase in pitcher *Threat* is associated with roughly 5% lower odds of a pickoff attempt ($\exp(-0.052) \approx 0.95$). After one prior pickoff attempt, the odds of attempting another pickoff increase by about 75% ($\exp(0.561) \approx 1.75$). After two prior pickoff attempts, the odds increase by about 65% ($\exp(0.502) \approx 1.65$). Relative to 0 outs, 1 out is associated with higher pickoff-attempt odds ($\exp(0.092) \approx 1.10$), while the 2-outs contrast is not statistically distinguishable from zero.

In the pickoff-success-given-attempt model (Table 4), a one-standard-deviation increase in lead distance is associated with roughly 186% higher odds of pickoff success ($\exp(1.050) \approx 2.86$), while a one-standard-deviation increase in pitcher *Threat* is associated with roughly 121% higher odds ($\exp(0.791) \approx 2.21$). After one prior pickoff attempt, the odds of a successful pickoff increase by about 68% ($\exp(0.518) \approx 1.68$). After two prior pickoff attempts, the effect is not statistically significant ($p = 0.112$). The runner-specific random effect shows substantial individual variation (variance = 0.444, SD = 0.666), suggesting that some runners are more vulnerable to being picked off after conditioning on the observed covariates.

In the steal-attempt-given-no-pickoff model (Table 5), a one-standard-deviation increase in *Threat* is associated with roughly 38% lower odds of a steal attempt ($\exp(-0.471) \approx 0.62$). Attempt odds are higher with slower catchers (about 6% per standard deviation; $\exp(0.060) \approx 1.06$) and faster runners (about 145% per standard deviation; $\exp(0.896) \approx 2.45$). After one prior pickoff attempt, steal-attempt odds increase by about 145% ($\exp(0.898) \approx 2.46$). After two prior pickoff attempts, attempt odds increase by about 290% ($\exp(1.360) \approx 3.90$). Outs is also a strong predictor in this stage: relative to 0 outs, attempt odds are about 37% higher at 1 out ($\exp(0.317) \approx 1.37$) and about 94% higher at 2 outs ($\exp(0.664) \approx 1.94$). Random effects show substantial runner heterogeneity (variance = 0.523, SD = 0.723) and smaller but nonzero catcher heterogeneity (variance = 0.014, SD = 0.117).

In the steal-success-given-attempt model (Table 6), a one-standard-deviation increase in lead distance is associated with roughly 16% higher odds of success ($\exp(0.148) \approx 1.16$), while a

one-standard-deviation increase in pitcher *Threat* is associated with roughly 25% lower odds ($\exp(-0.286) \approx 0.75$). Success odds are also higher with slower catcher pop times (about 33% per standard deviation; $\exp(0.286) \approx 1.33$). Sprint speed shows a marginal positive association ($p = 0.071$), disengagement-stage effects are not statistically significant, and the outs terms are not statistically significant in this stage. Random effects remain modest (runner variance = 0.127, catcher variance = 0.011), suggesting that much of the observed variation in steal success is captured by the included situational variables.

Overall, these associations align with baseball intuition and indicate that the sequential model captures the intended trade-off between lead size, pickoff risk, and steal success. The probability estimates should be interpreted as model-implied quantities for the analyzed pitch sample rather than as definitive causal effects.

3.2 Expected Runs and Lead Deviation

For each prediction context, consisting of the covariates $X_i = (\theta_i, p_i, v_i, r_i, o_i)$ and, where available, runner and catcher IDs, we compute $\text{xRuns}(L \mid X_i)$ for each candidate lead distance $L \in \mathcal{L}$ using Equation 6. We then report the model-implied optimal lead as in Equation 7:

$$L_i^* = \arg \max_{L \in \mathcal{L}} \text{xRuns}(L \mid X_i).$$

We define lead deviation as

$$d_i \equiv L_i^{\text{obs}} - L_i^*. \quad (8)$$

Positive values indicate observed leads that are larger than the model-implied optimum, while negative values indicate observed leads that are smaller. Across all runner-on-first observations, deviations are centered slightly above zero, with a mean of +0.13 ft (Figure 1). On steal attempts, the mean deviation remains positive at +0.10 ft (Figure 2). Thus, runners in this sample tend to take slightly more aggressive leads than the model-implied optimum, though the average difference is small relative to typical primary lead distances.

3.3 Case Study: Cubs vs. Pirates, August 28, 2024

In an August 28, 2024 game between the Chicago Cubs and Pittsburgh Pirates, Pete Crow-Armstrong (PCA) took an 11.46-foot lead against pitcher Paul Skenes and catcher Yasmani Grandal (Figure 3). Given PCA’s sprint speed of 30 ft/s, Skenes’s estimated Threat of 4.04, Grandal’s pop time of 2.09 seconds, and the game context, the model produces an optimal lead estimate of $L_i^* = 11.33$ ft. PCA’s observed lead exceeds this estimate by only 0.13 ft, so the example should be read primarily as an illustration of the model-implied trade-off rather than as evidence of a meaningfully suboptimal lead.

Figure 4 displays the model-implied probabilities of SB, CS, and PK as functions of primary lead L . Figure 5 separately shows that a lead distance of 11.33 ft maximizes the estimated expected run value (+0.012). This turning point reflects the lead-taking trade-off: larger leads increase the probability of reaching second safely, but eventually the increased pickoff risk dominates.

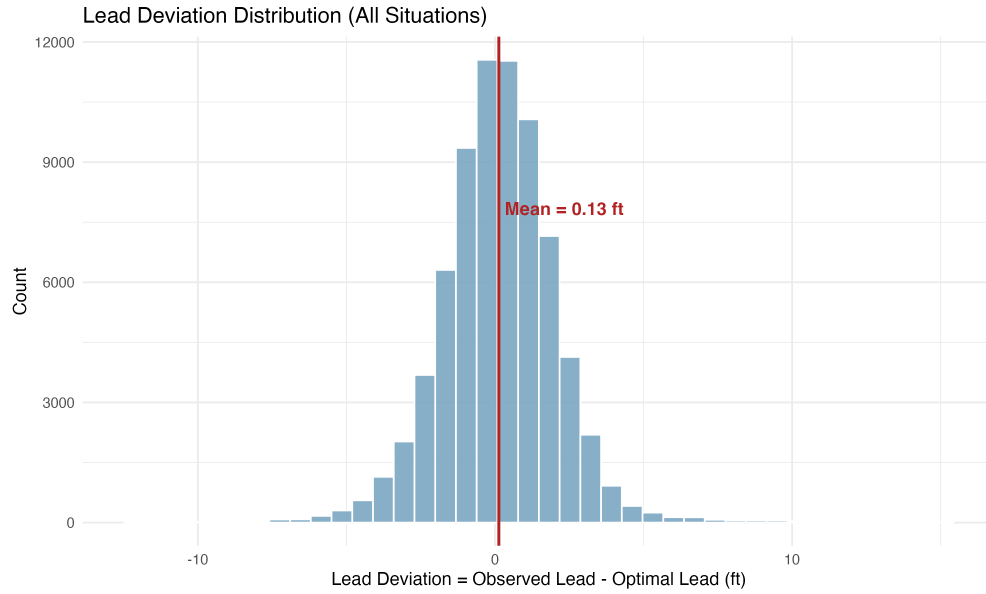


Figure 1: Distribution of lead deviations across all situations. Positive values indicate observed leads larger than the model-implied optimum; the red vertical line and label mark the sample mean (+0.13 ft).

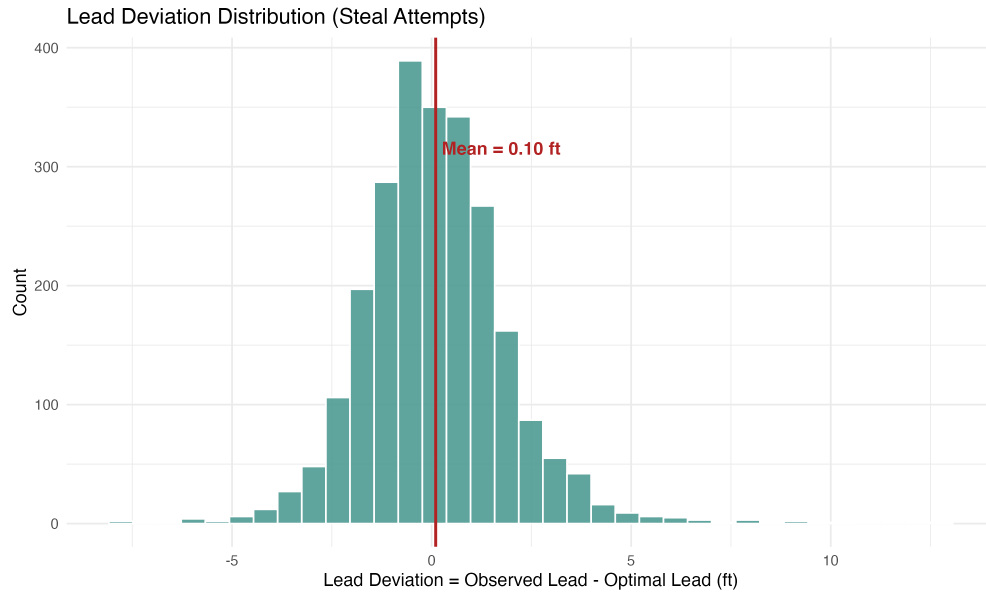


Figure 2: Distribution of lead deviations when a stolen base is attempted. The mean remains positive but is lower than the all-situations mean; the red vertical line and label mark the sample mean (+0.10 ft).



Figure 3: Pete Crow-Armstrong on first base, Paul Skenes pitching, and Yasmani Grandal catching offscreen. ([Marquee Sports Network, 2024](#))

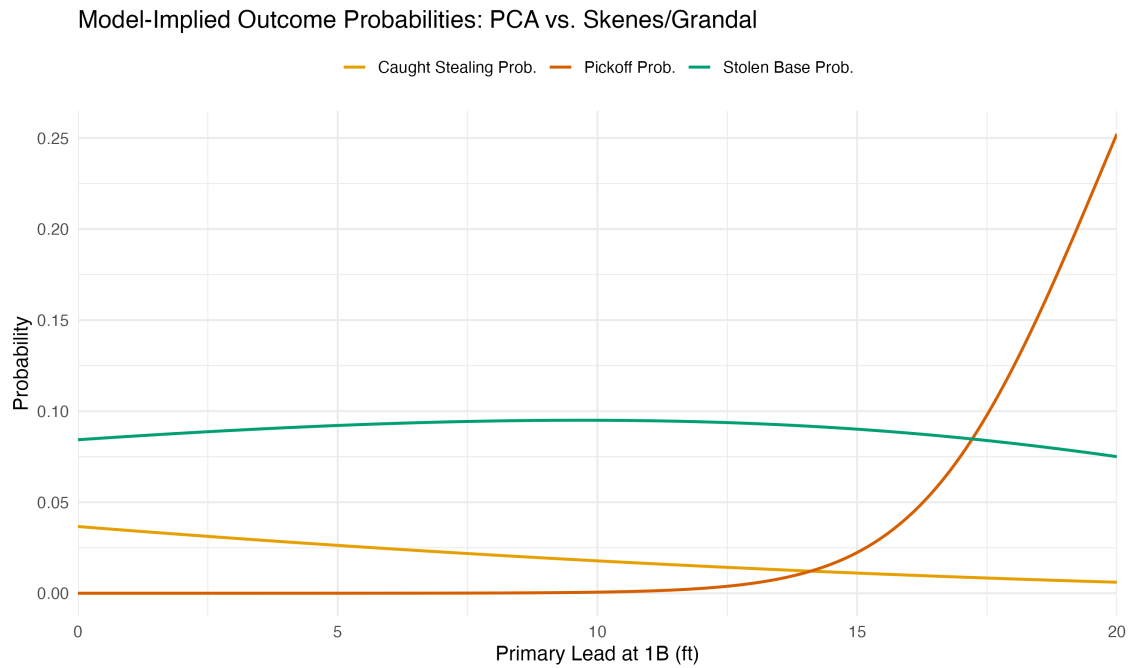


Figure 4: Model-implied outcome probabilities as functions of Pete Crow-Armstrong's primary lead distance.

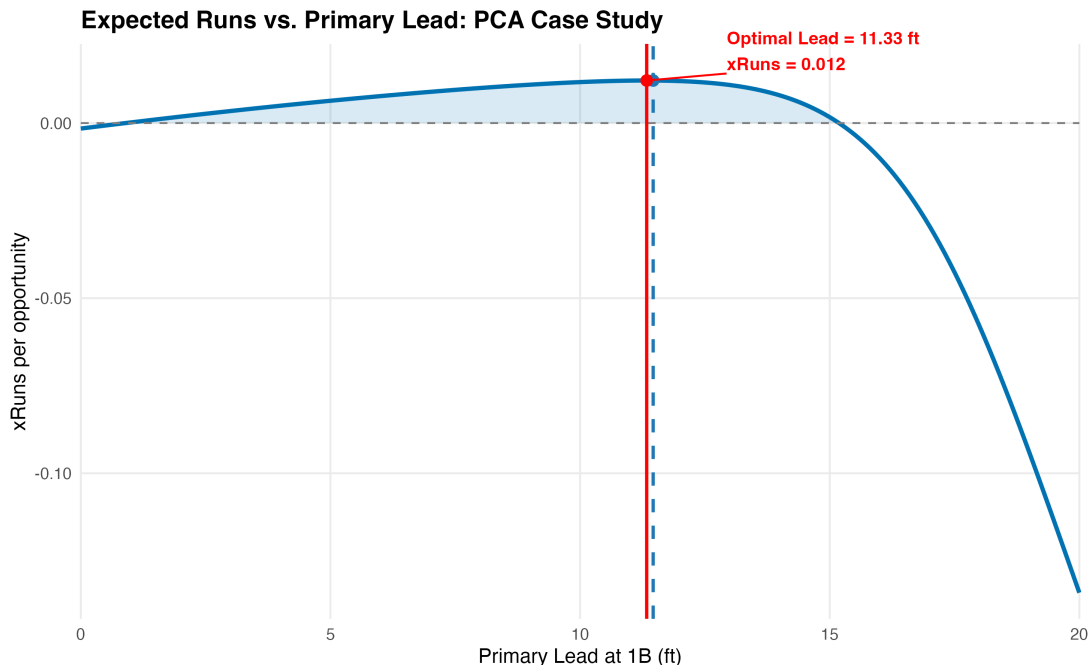


Figure 5: Estimated expected run value as a function of Pete Crow-Armstrong’s primary lead distance. The red marker denotes the model-implied optimal lead, and the blue dashed marker denotes PCA’s observed lead in the play.

3.4 Player and Team Lead Deviation

Lead deviation varies across teams (Figure 7). The New York Yankees exhibit the largest mean absolute deviation in our sample by a wide margin, about 0.61 ft above the next highest team, while the Cleveland Guardians and Seattle Mariners are closest to the model-implied optimum on average.

The player-level tables (Tables 7–9) provide context for this team-level pattern. Three Yankees runners (Anthony Volpe, Juan Soto, and Aaron Judge) appear among the ten most conservative runners, with mean deviations between -2.32 and -3.32 ft. This concentration is consistent with the Yankees’ high team-level absolute deviation. Across the broader sample, the ten most conservative runners deviate from the model-implied optimum by 1.9 to 3.6 ft on average, while the ten most aggressive deviate by 2.4 to 5.3 ft. The runners closest to the optimum show mean absolute deviations below 0.85 ft. All runner lists below include only players with at least 15 observations.

These plots should be read as exploratory summaries rather than definitive rankings, since the estimates do not include uncertainty intervals.

3.5 Run-Expectancy Cost

While lead deviation in feet describes how far observed leads sit from the model-implied optimum, it does not directly express the run-expectancy consequence of that deviation. We

define the model-implied run cost of an observed lead as

$$\text{xRuns_lost}_i \equiv \text{xRuns}(L_i^* | X_i) - \text{xRuns}(L_i^{\text{obs}} | X_i), \quad (9)$$

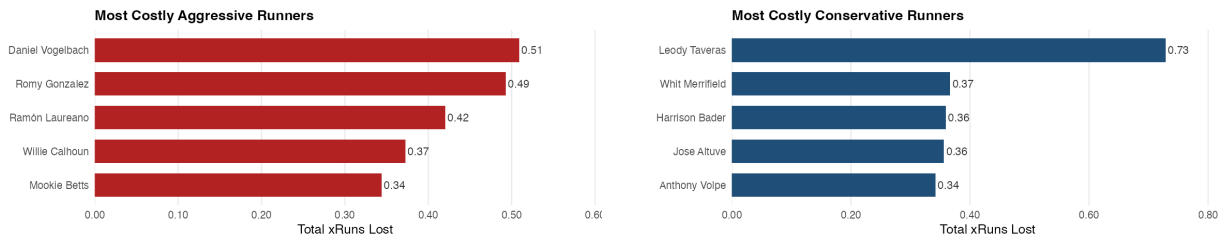
which is non-negative since L_i^* maximizes the expected run value at context X_i .

Across the analyzed 2024 sample, the model implies a total of 38.8 expected runs lost to suboptimal leads, with a mean of 0.00053 xRuns lost per opportunity.

Our analyzed data include observed pickoff attempts, stolen-base attempts on takes, and no-swing runner-on-first opportunities included in the 2024 pitch-level sample. Because pitches with swings can introduce additional advancement mechanisms not represented in Equation 6, we interpret the resulting 38.8 expected runs lost as a model-implied total over the analyzed steal–pickoff opportunity set, rather than as a claim about every runner-on-first pitch in the full 2024 season.

Aggregating xRuns_lost_i to the team level (Figure 8) shows that the Los Angeles Angels accumulated the largest total run cost in 2024 (2.00 xRuns), followed by the Boston Red Sox (1.79) and the Los Angeles Dodgers (1.72). The team ordering by run cost differs substantially from the ordering by mean absolute deviation (Figure 7). Most notably, the New York Yankees, who exhibit the largest mean absolute deviation in our sample, rank only 13th in total run cost. This contrast suggests that not all deviations are equally costly.

At the player level, ranking runners by total xRuns_lost_i produces different leaderboards depending on the direction of their typical deviation. Figure 6 presents the five most costly aggressive runners (positive mean deviation) and the five most costly conservative runners (negative mean deviation), each ranked by total run cost. Daniel Vogelbach leads the aggressive table at 0.51 xRuns lost despite only 25 observations, reflecting a high per-opportunity cost. On the conservative side, totals are more modest: the most costly conservative runner accumulates 0.73 xRuns over the season.



(a) Most costly aggressive runners (positive mean deviation).

(b) Most costly conservative runners (negative mean deviation).

Figure 6: Top five runners by total model-implied run cost in 2024, split by direction of mean lead deviation. Aggressive runners (left) reach higher per-opportunity costs; conservative runners (right) accumulate more modest totals.

The contrast between the aggressive and conservative tables reflects a broader pattern: aggressive deviations carry a substantially higher run cost per opportunity than conservative deviations. Across all observations, the mean per-opportunity run cost is 0.00079 xRuns on

pitches with a positive deviation and 0.00024 xRuns on pitches with a negative deviation, a ratio of approximately 3.3 to 1. This asymmetry explains the team-level contrast noted earlier: the New York Yankees deviate most in absolute terms but rank only 13th in total run cost because their deviation is concentrated on the conservative side, which the model treats as a less costly form of suboptimal lead choice. The implication is that lead deviation and run cost are not interchangeable measures of suboptimal baserunning, and that evaluating runners on deviation alone overstates the cost of conservative behavior relative to aggressive behavior.

4 Discussion

4.1 Conclusions

This study estimates primary lead distances off first base by placing the steal–pickoff trade-off on an outs-specific run-expectancy scale. The sequential model combines pickoff, steal-attempt, and steal-success probabilities with RE24 weights to produce a context-specific model-implied optimal lead and a run-cost measure for observed deviations.

In the analyzed 2024 sample, observed leads are close to the model-implied optimum on average: +0.13 ft across all runner-on-first opportunities and +0.10 ft on steal attempts. The fitted trade-off produces an expected-run profile with an interior maximum in the PCA case study and in many of the empirical contexts we evaluate. The associated model-implied cost is 38.8 expected runs, but that cost is asymmetric: aggressive deviations carry roughly 3.3 times the per-opportunity run cost of conservative deviations.

Lead deviation in feet and run cost in expected runs are therefore not interchangeable measures of baserunning quality. The framework provides a compact way to compare lead-taking behavior across players and teams while preserving the expected-runs consequences of the underlying baseball trade-off.

4.2 Limitations and Future Directions

The analysis has several limitations. First, we do not observe steal intent on a given pitch. Lead distance may therefore proxy for unobserved steal decisions, especially on pitches where the runner was planning to go. We partially address this by excluding lead distance from the steal-attempt model, but direct intent labels would allow a cleaner separation between strategic intent and realized positioning.

Second, the fitted probabilities should be interpreted as model-implied quantities for the analyzed pitch sample rather than as fully calibrated league-wide event probabilities. This point mainly affects the level of the expected run value function, not the qualitative logic of the trade-off, but it is important when interpreting the numerical optimum as a recommendation.

Third, the disengagement variable is constructed from observed pickoff attempts only. Because step-offs are not observed, this variable may understate the true number of disengage-

ments in a plate appearance.

Fourth, our model considers only the steal–pickoff trade-off (SB, CS, and PK). A larger lead also affects a runner’s ability to advance on balls in play, including reaching second on a single, scoring from first on a double, and avoiding double plays. These effects are outside the scope of our model and are not captured in our run-cost estimates.

Fifth, we restrict attention to the base state with a runner on first, with second and third base empty. Other base states would change both the RE24 payoffs and defensive behavior. Extending the model to all base-out states is a natural next step.

Finally, the current model treats pitches as conditionally independent. In practice, pitchers, runners, and catchers adapt within plate appearances and across games. A dynamic model with plate-appearance history, pitch type, batter handedness, pitch location, score, inning, and leverage could better capture this strategic interaction.

4.3 Reproducibility

All analysis code and figure scripts are available at https://github.com/WhartonSABI/mba2025_base-stealers. Proprietary MLB data are not publicly available.

5 Acknowledgments

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References

- Baseball-Reference (n.d.). Major league baseball 1950 season summary. Sports Reference LLC.
- Baseball Savant (n.d.). Base stealing run value leaderboard. Major League Baseball Advanced Media.
- Brent, R. P. (1973). *Algorithms for Minimization without Derivatives*. Prentice-Hall.
- FanGraphs (n.d.). Re24 (run expectancy based on 24 base/out states). FanGraphs Library.
- James, B. (2023). Bill james handbook excerpt: Baserunning in its own self. Sports Info Solutions.
- Kagan, D. (2013). Stolen base physics. *The Physics Teacher*, 51(5):269–271.
- Lindbergh, B. (2015). Statcast baserunning data and ichiro suzuki’s secrets. Grantland.

- Marquee Sports Network (2024). Chicago cubs vs. pittsburgh pirates [tv broadcast]. Television broadcast, Marquee Sports Network, August 28, 2024.
- McMurray, J. (2015). Stolen bases in the deadball era: A relentless approach. Society for American Baseball Research.
- Powers, S. (2026). Lead distance under a pickoff limit in major league baseball: A sequential game model. *arXiv*.
- Turocy, T. L. (2014). The economics of theft: An analysis of base stealing in baseball as a two-player game. Working paper.
- Vazzana, A. (2016). Maury wills and the value of a stolen base. Society for American Baseball Research.

Appendix

A Full Model Estimates

Table 3: Logistic regression for pickoff attempts (PO)

Fixed Effects				
Term	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.3242	0.0244	-95.142	$< 2 \times 10^{-16}$
lead_scaled	0.3089	0.0121	25.477	$< 2 \times 10^{-16}$
threat_scaled	-0.0521	0.0118	-4.411	1.03×10^{-5}
dis_state1	0.5614	0.0343	16.366	$< 2 \times 10^{-16}$
dis_state2	0.5017	0.0730	6.872	6.33×10^{-12}
outs_f1	0.0920	0.0311	2.963	3.05×10^{-3}
outs_f2	-0.0192	0.0316	-0.606	5.44×10^{-1}
Random Effects				
No random effects (fixed-effects GLM)				
Number of obs: 72,645				

Table 4: Mixed-effects logistic regression for pickoff success given attempt (PK | PO)

Fixed Effects				
Term	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-5.2758	0.2266	-23.278	$< 2 \times 10^{-16}$
lead_scaled	1.0495	0.1125	9.327	$< 2 \times 10^{-16}$
threat_scaled	0.7911	0.1198	6.602	4.07×10^{-11}
dis_state1	0.5181	0.2230	2.324	2.01×10^{-2}
dis_state2	0.6644	0.4176	1.591	1.12×10^{-1}
Random Effects				
Runner1B_ID (Intercept): Variance = 0.4440, SD = 0.6663				
Number of obs: 7,439; groups (Runner1B_ID): 530				

Table 5: Mixed-effects logistic regression for steal attempt given no pickoff (ATT | $\neg$ PO)

Fixed Effects				
Term	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-4.4160	0.0667	-66.223	$< 2 \times 10^{-16}$
threat_scaled	-0.4712	0.0173	-27.323	$< 2 \times 10^{-16}$
poptime_scaled	0.0600	0.0274	2.192	2.84×10^{-2}
sprint_scaled	0.8957	0.0486	18.424	$< 2 \times 10^{-16}$
dis_state1	0.8983	0.0558	16.096	$< 2 \times 10^{-16}$
dis_state2	1.3603	0.0984	13.822	$< 2 \times 10^{-16}$
outs_f1	0.3169	0.0613	5.170	2.34×10^{-7}
outs_f2	0.6642	0.0592	11.221	$< 2 \times 10^{-16}$
Random Effects				
Runner1B_ID (Intercept): Variance = 0.5233, SD = 0.7234				
CatcherID (Intercept): Variance = 0.0136, SD = 0.1165				
Number of obs: 65,206; groups (Runner1B_ID): 602; CatcherID: 83				

Table 6: Mixed-effects logistic regression for steal success given attempt (SB | ATT)

Fixed Effects				
Term	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.8672	0.1320	6.572	4.96×10^{-11}
lead_scaled	0.1480	0.0652	2.268	2.33×10^{-2}
threat_scaled	-0.2861	0.0493	-5.808	6.34×10^{-9}
poptime_scaled	0.2856	0.0545	5.242	1.59×10^{-7}
sprint_scaled	0.1175	0.0651	1.805	7.11×10^{-2}
dis_state1	-0.1404	0.1209	-1.161	2.46×10^{-1}
dis_state2	-0.1608	0.2026	-0.794	4.27×10^{-1}
outs_f1	0.0261	0.1359	0.192	8.48×10^{-1}
outs_f2	0.1899	0.1323	1.435	1.51×10^{-1}
Random Effects				
Runner1B_ID (Intercept): Variance = 0.1272, SD = 0.3567				
CatcherID (Intercept): Variance = 0.0108, SD = 0.1039				
Number of obs: 2,434; groups (Runner1B_ID): 417; CatcherID: 83				

B Team and Player Lead Deviation and Run Cost Figures

This appendix presents the team- and player-level summaries of lead deviation and run cost referenced in Sections 3.4 and 3.5. All runner lists include only players with at least 15 observations.

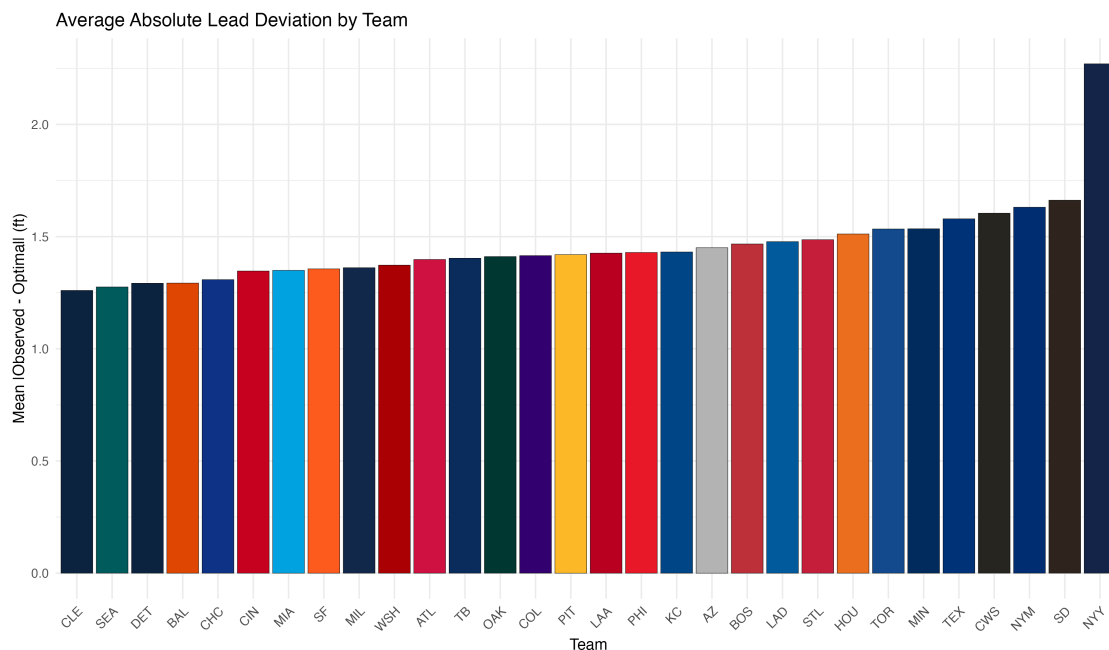


Figure 7: Average absolute lead deviation by team. Lower values indicate observed leads that are closer to the model-implied optimum on average.

Table 7: Baserunners with the smallest absolute average lead deviation among runners with at least 15 observations.

Runner	Team	Absolute Deviation (ft)	Observations
Jack López	LAA	0.55	38
David Dahl	PHI	0.56	22
Braden Shewmake	CWS	0.57	19
Drew Millas	WSH	0.59	26
Evan Carter	TEX	0.72	33
Eli White	ATL	0.72	27
Niko Kavadas	LAA	0.74	58
Brice Turang	MIL	0.80	331
Christian Encarnacion-Strand	CIN	0.81	48
Eric Haase	MIL	0.83	17

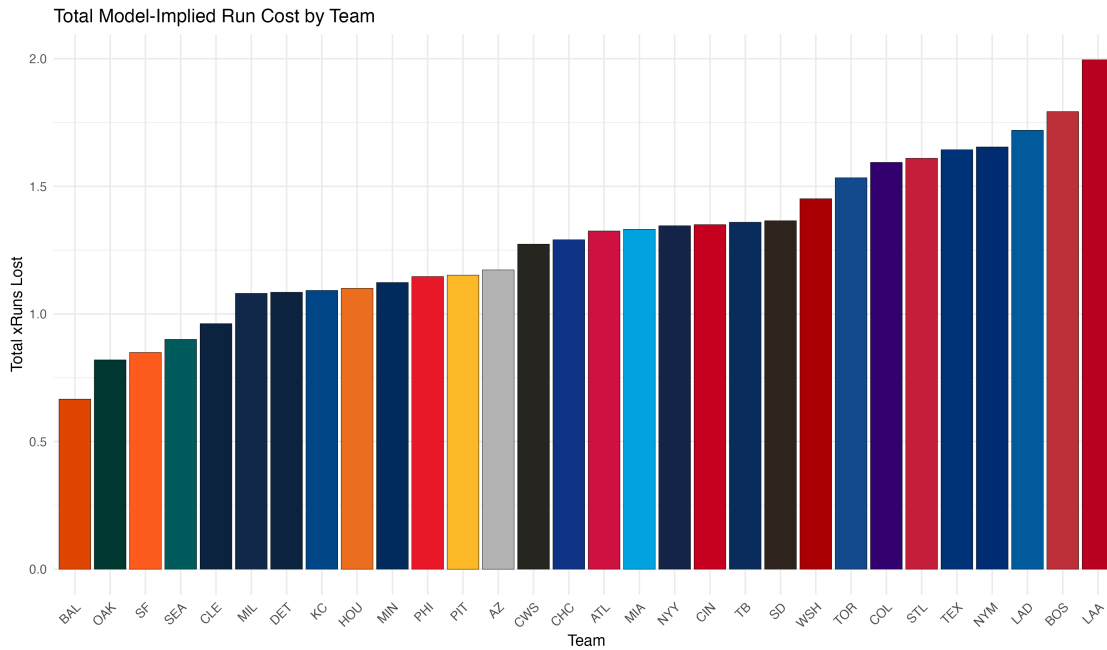


Figure 8: Total model-implied run cost by team in 2024. Compare to Figure 7: teams that deviate most in feet do not necessarily incur the largest run cost.

Table 8: Baserunners with the most positive average lead deviations among runners with at least 15 observations.

Runner	Team	Mean Deviation (ft)	Observations
Daniel Vogelbach	TOR	5.29	25
Wilmer Flores	SF	2.94	89
Brandon Crawford	STL	2.77	30
Alan Trejo	COL	2.57	25
Eloy Jiménez	CWS/BAL	2.57	103
Cade Marlowe	SEA	2.52	18
Kris Bryant	COL	2.49	64
Willie Calhoun	LAA	2.49	92
Jacob Stallings	COL	2.45	95
Enrique Hernández	LAD	2.40	162

Table 9: Baserunners with the most negative average lead deviations among runners with at least 15 observations.

Runner	Team	Mean Deviation (ft)	Observations
Brooks Baldwin	CWS	-3.58	30
Anthony Volpe	NYY	-3.32	356
Juan Soto	NYY	-2.76	448
Aaron Judge	NYY	-2.32	340
Francisco Alvarez	NYM	-2.09	97
Josh Lowe	TB	-2.08	132
Bobby Dalbec	BOS	-1.95	23
Carlos Correa	MIN	-1.90	156
Ceddanne Rafaela	BOS	-1.90	160
Harrison Bader	NYM	-1.85	169